Appendix C.1. Rank-One Determinant Route: Determinant Algebra with the $\frac{2}{3}$ Normalization

Consider the rank-one determinant route Lagrangian

$$\mathcal{L}_{\rm ROD} = \sqrt{-g}\Big(\sqrt{\det\big[\mathbf{1} + \tfrac{2}{3}\lambda\,\tau\,\mathcal{T}\big]} - 1\Big), \qquad \lambda \in \mathbb{R}. \tag{A31}$$

Using $\sqrt{\det(\mathbf{1}+X)} = 1 + \frac{1}{2}\mathrm{Tr}X + \frac{1}{8}[(\mathrm{Tr}X)^2 - 2\mathrm{Tr}X^2] + \mathcal{O}(X^3)$ and $\mathrm{Tr}\mathcal{T} = 0$, the quadratic piece is

$$\begin{aligned}\delta^2\mathcal{L}_{\rm ROD} &= \sqrt{-g}\,\frac{1}{8}\Big[-2\,\mathrm{Tr}\Big((\tfrac{2}{3}\lambda\,\tau)^2\mathcal{T}^2\Big)\Big] = -\frac{1}{4}\big(\tfrac{2}{3}\big)^2\lambda^2\,\sqrt{-g}\,\tau^2\,\mathrm{Tr}(\mathcal{T}^2)\\ &= -\frac{1}{4}\cdot\frac{4}{9}\cdot\frac{3}{4}\,\lambda^2\,\sqrt{-g}\,\tau^2 = -\frac{1}{12}\,\lambda^2\,\sqrt{-g}\,\tau^2.\end{aligned} \tag{A32}$$

With $\tau^2 = 9\eta^2|\Sigma_\epsilon|$ and $\mathrm{sgn}(\Sigma_\epsilon)\,I_T = -6\eta^2|\Sigma_\epsilon|$,

$$-\frac{1}{12}\lambda^2\tau^2 = -\frac{1}{12}\lambda^2\,(9\eta^2|\Sigma_\epsilon|) = -\frac{3}{4}\lambda^2\eta^2|\Sigma_\epsilon| = \Big(\frac{\lambda^2}{8}\Big)\,\mathrm{sgn}(\Sigma_\epsilon)\,I_T,$$

so that

$$\boxed{\delta^2\mathcal{L}_{\rm ROD} = A_\star\,\sqrt{-g}\,\mathrm{sgn}(\Sigma_\epsilon)\,I_T \;+\; \nabla_\mu J^\mu_{\rm ROD}, \qquad A_\star = \frac{\lambda^2}{8}.} \tag{A33}$$

At the bulk-density level, one may take $J^\mu_{\rm ROD} \equiv 0$. For unified boundary diagnostics, we adopt a common canonical representative $J^\mu_{\rm can}$ for *all* three routes (Appendix C.4).

Appendix C.2. Closed-Metric Route: Rank-One Deformation Equals Rank-One Determinant Route to $\mathcal{O}(\mathcal{T}^2)$

Take the rank–one deformation

$$\tilde{g}_{\mu\nu} = g_{\mu\nu} + \tfrac{2}{3}\lambda\,\tau\,\hat{n}_\mu\hat{n}_\nu,$$

so that

$$\sqrt{-\tilde{g}} = \sqrt{-g}\,\sqrt{\det\Big[\mathbf{1} + \tfrac{2}{3}\lambda\,\tau\,\mathcal{T}\Big]}.$$

Define

$$\mathcal{L}_{\rm CM} \equiv \sqrt{-\tilde{g}} - \sqrt{-g}.$$

The same algebra gives

$$\boxed{\delta^2\mathcal{L}_{\rm CM} = A_\star\,\sqrt{-g}\,\mathrm{sgn}(\Sigma_\epsilon)\,I_T \;+\; \nabla_\mu J^\mu_{\rm CM}, \qquad A_\star = \frac{\lambda^2}{8}.} \tag{A34}$$

Thus rank-one determinant route and CM have the *same bulk* coefficient $A_\star$ and differ only by improvements.

Appendix C.3. PT-Even CS/Nieh–Yan Shadow: Quadratic Reduction (σ_ϵ)

Using $T^A \wedge T_A = d(e^A \wedge T_A) - e^A \wedge e^B \wedge R_{AB}$, applying $*$ and the scalar PT projector (A2), and evaluating after C1, the PT-even piece reduces at quadratic order to

$$\boxed{\delta^2\mathcal{L}_{\rm CS^+} = A_\star\,\sqrt{-g}\,\mathrm{sgn}(\Sigma_\epsilon)\,I_T \;+\; \nabla_\mu J^\mu_{\rm CS^+}, \qquad A_\star = \frac{\lambda^2}{8},} \tag{A35}$$

where the Nieh–Yan assignment (A5) reshuffles only boundary conventions inside the PT–even sector.